

VEVAN O NARAIN

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Third-year B.Tech CSE (AI/ML) student at VIT Vellore with a 9.44 GPA, seeking an SDE internship, skilled in full-stack development and machine learning. Experienced building production-grade systems with React, Node.js, Docker, and PyTorch, with a focus on real-world impact across NLP, computer vision, and collaborative tooling.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, Java, C++, C
- **Frameworks & Libraries:** React.js, Node.js, Express.js, TensorFlow, Keras, PyTorch, Scikit-learn
- **Databases:** PostgreSQL, MongoDB, Redis
- **Developer Tools:** Git, GitHub, Docker, VS Code, Jupyter Notebook, Google Colab, Streamlit
- **Core Concepts:** Data Structures & Algorithms, Object-Oriented Programming, REST APIs, System Design

PROJECTS

Vync - Real-Time Collaborative Editor [*React, Node.js, Socket.IO, Yjs, PostgreSQL, Redis, Docker*] [GitHub](#)

- Built a Google Docs-style collaborative code editor supporting real-time multi-user editing using CRDTs (Yjs) and WebSockets with support for six programming languages (JavaScript, Python, TypeScript, Java, C++, and Go).
- Implemented conflict-free synchronization of code changes across multiple clients with low-latency updates and automatic state reconciliation.
- Developed sandboxed multi-language code execution using Docker containers with CPU and memory limits to ensure secure and isolated execution environments.

10Kwery - SEC Filing RAG Copilot [*Python, LangChain, ChromaDB, BM25, Cross-Encoder, Claude API, FastAPI*] [GitHub](#)

- Built a retrieval-augmented generation system over SEC 10-K/10-Q filings and earnings transcripts, enabling natural-language Q&A with citation-grounded answers traceable to exact filing sections.
- Designed a hybrid retrieval pipeline combining dense vector search with BM25 keyword matching and cross-encoder reranking, improving citation accuracy over naive semantic search.
- Implemented section-aware document chunking that preserves financial table structure and metadata (company, fiscal period, filing section), reducing numerical hallucination in generated answers.

Plant Leaf Disease Detection [*Python, TensorFlow, Keras, ResNet50, Streamlit, Scikit-learn, Pillow, Kaggle API*] [GitHub](#)

- Achieved 98.75% accuracy in detecting 38 plant diseases by fine-tuning ResNet50 on a dataset containing over 50,000 plant leaf images and applied transfer learning and image preprocessing techniques to improve model generalization across multiple crop species and disease classes.
- Deployed an interactive Streamlit application that enables real-time image upload and disease classification with prediction confidence scores.

Real-Time Emotion Detector [*Python, TensorFlow/Keras, OpenCV, CNN, NumPy, Pandas*] [GitHub](#)

- Developed a real-time facial emotion recognition system using OpenCV for face detection and a CNN trained on the FER-2013 dataset containing over 30,000 facial images.
- Classified seven facial emotions from live webcam input, enabling continuous real-time emotion prediction and visualization.
- Implemented image preprocessing and data augmentation to improve model robustness across varying conditions.

EDUCATION

• **B.Tech in Computer Science and Engineering (AI & ML Specialization)**

Vellore Institute of Technology, Vellore

Cumulative GPA: 9.44 | Expected Graduation: May, 2028

• **Senior Secondary (Class XII)** — Modern School Barakhamba Road, CBSE Board

Percentage: 90.2% (PCM with CS) | Completed: May, 2024

• **Secondary (Class X)** — Modern School Barakhamba Road, CBSE Board

Percentage: 96% | Completed: May, 2022

CERTIFICATIONS & ACHIEVEMENTS

- **Machine Learning A-Z** by SuperDataScience Team
- **The MERN Fullstack Guide** by Maximilian Schwarzmüller and Manuel Lorenz
- Solved 200+ problems on LeetCode, strengthening problem-solving skills and proficiency in data structures and algorithms.